

# Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau

## Phase 2 – Requirement Analysis

### 1. Customer Journey Map

The journey below traces how a stakeholder (e.g., a Brand Marketing Manager) moves from discovering a problem to acting on it, mapped against the three business scenarios.

**Awareness:** Receives an alert or notices a decline in interest for certain products/ingredients (Scenario 1: Monitoring Consumer Preferences).

**Investigation:** Opens the Cosmetic Insights dashboard to assess the extent and impact of the trend, filtering by brand, category, price, and skin type.

**Diagnosis:** Identifies whether the issue is linked to negative reviews or safety concerns for specific products (Scenario 2: Addressing Product Concerns).

**Action:** Deploys interventions — targeted campaigns, formulation changes, recalls, or transparent communication — based on the insights surfaced.

**Anticipation:** Uses predictive analytics and historical trend data to proactively innovate new products and stay ahead of competitors (Scenario 3: Predictive Analysis and Product Innovation).

### 2. Data Flow Diagram

Data Source (Excel/CSV – brand, category, price, rank, skin type) → Tableau Data Connection & Cleaning → Calculated Fields & Data Preparation → Tableau Visualizations → Published Dashboard & Story → Embedded in Flask Web App → End User (Brand Manager / Analyst / Regulatory Body).

### 3. Solution Requirements

#### Functional Requirements

- Connect and clean the cosmetics dataset (brand, category, price, rank, skin type).
- Provide interactive filters for Brand, Category, Skin Type, and Price Range.
- Visualize brand performance, pricing patterns, product distribution, and skin-type coverage.
- Provide a multi-scene story walking through the three business scenarios.
- Embed the published dashboard and story into a Flask-based web application.

#### Non-Functional Requirements

- Dashboard should load and respond quickly even with the full dataset applied.
- Layout should be responsive across different screen sizes (device-specific layouts).

- Visuals should be intuitive and require minimal training for business users.
- System should support stable performance with standard hardware (see System Requirements).

## **4. Technology Stack**

- Data Source: Excel / CSV (cosmetics dataset)
- Analytics & Visualization: Tableau Desktop / Tableau Public
- Web Framework: Flask (Python) for embedding dashboard and story
- Frontend: HTML, CSS for the web UI hosting the embedded dashboard
- Operating System: Windows 10/11 (64-bit) or macOS (latest supported versions)
- Browser: Latest version of Chrome, Edge, or Safari

This completes Phase 2 (Requirement Analysis), covering: Customer Journey Map, Data Flow Diagram, Solution Requirements, and Technology Stack.